

Renewable-Energy PoC - Executive Summary

Chilean renewable-energy data platform on Databricks | updated 2026-07-17

Overview

A medallion (bronze -> silver -> gold) lakehouse on Databricks. It ingests real data from four independent sources, turns it into analytics-ready metrics across three domains, and is fully governed and monitored - all in dev, on isolated branches; production is the project lead's.

Four data sources - all on real data

- Weather: NASA POWER (historical hourly) + DMC (real-time measured) + MinEnergia/DGF (modeled).
- Resource (generation): CEN / Coordinador - PUBLIC by law (Art. 72-8, Ley 20.936), no cost, no registration.
- Conglomerate (country): Our World in Data (OWID), open CSV.

Key unblocks this cycle

- CEN generation was assumed to need a registered SIPUB account - it is public by law and reachable with the public key the Coordinador's own site exposes.
- Historical weather with the same fields as the DMC network is served cleanly by NASA POWER: a 10-year hourly full load (876,720 rows) is loaded in dev.

Results (verified in dev)

- Medallion runs end-to-end for all three domains.
- Weather: 10 plants ranked; 10-year historical hourly loaded, physically validated (Atacama = highest irradiance).
- Resource: real per-plant generation (coordinado / real / reducciones).
- Conglomerate: Chile-vs-World renewables lead grew +19 -> +38 (2017-2024).

Governance & data quality (Databricks-native)

- Data contracts per medallion step + DLT expectations; data glossary; Unity Catalog lineage + comments.
- Observability: three Lakehouse Data-Quality Monitors on gold (profiling + drift, auto dashboards) + a SQL alert.

Verify the medallion & jobs

- Resource pipeline: <https://dbc-54b27bae-2e91.cloud.databricks.com/pipelines/9bcfc32a-19f0-4327-b5bc-e8ee1cc4a3a4?o=7474645896934260>
- Conglomerate pipeline: <https://dbc-54b27bae-2e91.cloud.databricks.com/pipelines/b5d63282-a439-42ef-8e3b-cd707a9d5f58?o=7474645896934260>
- Weather pipeline: <https://dbc-54b27bae-2e91.cloud.databricks.com/pipelines/c108b287-98cc-43d6-86d6-5b63a9e6b4ae?o=7474645896934260>
- Weather job: <https://dbc-54b27bae-2e91.cloud.databricks.com/jobs/503330163541320?o=7474645896934260>
- Quality monitor (Resource gold): <https://dbc-54b27bae-2e91.cloud.databricks.com/sql/dashboardsv3/01f1809adb1a199384b80e96c469a206?o=7474645896934260>
- Pull requests: https://github.com/oxiboy/poc_databricks_evaluateserve/pulls